11 V-I CHARACTERISTICS OF ZENER DIODE

Aim:

To study the V-I characteristics of zener diode and its break down voltage.

Apparatus / Components required:

S. No	Name of the apparatus / Components	Range / Value	Quantity
1.	Regulated power supply	0-30 V DC, 2A	1 No.
2.	Multimeter	-	2 No.
3.	Zener diode	6.3V	1 No.
4.	Resistor	1kΩ, ½ W	1 No.
5.	Bread board	-	1 No.
6.	Single strand wire	-	Required quantity

Circuit Diagram

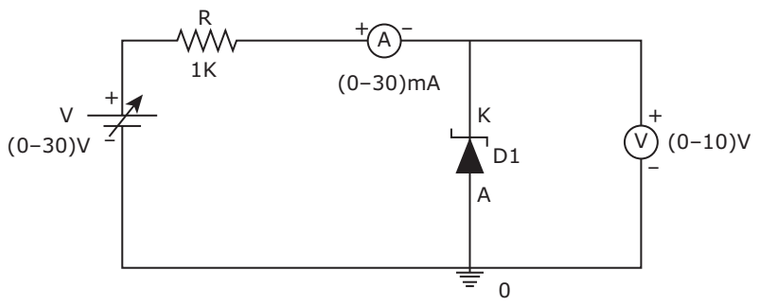


Fig 11.1 Regulator circuit

Procedure:

1. Connect the circuit as per the circuit diagram using the breadboard.
2. Vary the supply voltage in such a way that the readings are taken in steps of 0.5V.
3. Note down the corresponding ammeter readings (I_R).
4. Find out the reverse bias voltage (V_R).
5. Plot a graph between V_R & I_R.
6. Tabulate the values of V_R & I_R.

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